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Tutorial 6: Design a digital IIR low-pass filter for a specified cutoff frequency and realize an IIR filter in direct, cascade, and parallel forms.

1. Convert the analog filter into a digital filter by using the impulse-invariant technique with $T = 1$ s whose system function $H(s)$ is given by:

a) $H(s) = \frac{s+0.2}{(s+0.2)^2+9}$

b) $H(s) = \frac{1}{(s+1)(s+2)}$

c) $H(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$

2. Apply bilinear transformation to given $H(s)$ with $T=1$ s and find $H(z)$:

a) $H(s) = \frac{s^2+4.525}{s^2+0.692s+0.504}$

b) $H(s) = \frac{2}{(s+1)(s+3)}$

3. Convert the analog filter with system function

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

into a digital IIR filter using bilinear transformation. The digital filter

should have a resonant frequency of $\omega_r = \frac{\pi}{4}$

- 4 Obtain the direct Form I, direct Form II, cascade, and parallel realizations of the following IIR system :

a) $y(n) = -0.1y(n-1) + 0.2y(n-2) + 3x(n) + 3.6x(n-1) + 0.6x(n-2)$

b) $H(z) = \frac{3(2z^2+5z+4)}{(2z+1)(z+2)}$

c) $H(z) = \frac{0.28z^2+0.319z+0.04}{0.5z^3+0.3z^2+0.17z-0.2}$
